

DEBIDUTTA BEHERA

Software Engineer

Athagarh, Cuttack, Odisha, 754029, India

+91 7978213833 | debidutta.db@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Results-driven Software Engineer with hands-on experience designing, developing, and maintaining scalable software applications using Python, FastAPI, Django REST Framework, and React.js. Proven success collaborating with cross-functional teams to define and implement new features, while testing, debugging, and optimizing systems for performance, scalability, and reliability. Delivered 99.8% system uptime at DRDO's Integrated Test Range and consistently improved quality through automation and performance tuning. Eager to bring strong ownership and continuous improvement to a modern product engineering team.

EDUCATION

Radhakrishna Institute of Technology and Engineering B.Tech in Computer Science (74.37%)	2017 - 2021
Stewart Science College +2 in Science (52%)	2014 - 2017
S. S. V. M Tulasipur Matriculation (57%)	2013 - 2014

WORK EXPERIENCE

Apprentice Trainee — Software Developer & Automation Engineer	01/2025 - 01/2026
Integrated Test Range (ITR), DRDO Gov. Of India Chandipur, India	

- Built and maintained Python-based backend services for mission-critical enterprise applications, achieving 99.8% system uptime through robust service design and ownership of the full feature lifecycle.
- Automated UI testing protocols using Python Selenium and PyTest, reducing manual testing effort by 40% and deployment errors by 25% while improving release reliability.
- Optimized frontend/backend Python scripts to eliminate runtime bottlenecks, delivering a 35% reduction in execution time through proactive performance profiling and debugging.
- Participated in code reviews, authored technical documentation, and collaborated cross-functionally on internal software modules to identify improvements and implement new features.

PROJECTS

Cinema Seat Reservation System

- Python, Django REST Framework, React.js, TypeScript, PostgreSQL
- Architected RESTful backend endpoints with Django REST Framework to handle real-time booking transactions, seat-tier pricing logic, and data persistence in a PostgreSQL relational database.
- Built a dynamic, component-driven frontend using React.js and TypeScript with cross-device compatibility, real-time seat selection state management, and responsive UI rendering.
- Designed PostgreSQL schema with normalized tables for users, bookings, and pricing tiers, optimizing query performance for concurrent seat reservation transactions.

Digital Attendance Tracker

- Python, Django, React.js, TypeScript, PostgreSQL, MySQL, HTML, CSS
- Developed a full-stack web portal using Django backend and React.js frontend with TypeScript, enabling role-based access control for administrators and employees.

- Implemented secure authentication, responsive form components, and a PostgreSQL/MySQL database schema that reduced manual attendance errors and streamlined HR operations.

AWS Agreement RAG Chatbot

- FastAPI, Streamlit, LangChain
- Developed an AI-powered chatbot for querying AWS agreement documents using natural language.
- Built a FastAPI backend for document processing, embeddings, and retrieval-based question answering.
- Created a Streamlit frontend for document upload and interactive chat.
- Integrated LangChain and vector search to deliver accurate, context-aware responses.

AI-Powered Personal Assistant (JARVIS)

- Python, SpeechRecognition, NLP
- Built a voice-activated assistant using Python NLP libraries to parse and execute system commands, implementing robust error-handling that improved command recognition accuracy by 20%.

E-Commerce Frontend Web Application

- React.js, TypeScript, HTML5, CSS3
- Built modern UI pages including product listing, cart, and checkout workflows.
- Implemented component-based architecture for scalable frontend development.
- Used API integration and async/await for product data fetching.
- Ensured cross-browser compatibility and mobile responsiveness.

Automated Software Testing Framework

- Python, Selenium, PyTest
- Engineered a cross-browser automated UI testing framework validating login workflows, user interactions, and form functionality across Chrome, Firefox, and Edge.
- Generated structured HTML test reports with automated screenshot capture on failure, reducing average debugging time significantly and enabling faster CI/CD feedback loops.

ACHIEVEMENTS

- Strong understanding of frontend and backend fundamentals including component lifecycle, state management, and API handling.
- Built and deployed multiple frontend and full-stack projects using React and Python technologies.
- Continuously learning advanced frontend and backend engineering concepts and modern development practices.

SKILLS

Python, FastAPI, Django REST Framework, RESTful APIs, React.js, JavaScript, TypeScript, HTML5, CSS3, Bootstrap, Tailwind CSS, PostgreSQL, MongoDB, Selenium, PyTest, Software Development, Debugging, Performance Optimization, Scalability, Reliability, Cross-functional Collaboration, Documentation, Object-Oriented Programming(OOP), JSON, Git, GitHub, Strimlit, Vite, Adobe Photoshop, Figma, Blender, Canva, Problem Solving, Team Managment, CI/CD Principles, Cross-Browser UI, Responsive Design, OR-Tools CP-SAT, UI/UX, schema design, joins, indexing, PHP, C++, SQL, JAVA, Node.JS, JQuery, HTTP protocols, status codes, request/response structures, No-SQL, Collaboration, Digital & AI Mindset, Technical Communication, logic, coding, system changes, AI, ML, Flask, .net, OpenAI APIs, LangChain, CrewAI, n8n, Prompt Engineering, AI Agent Development, SDLE, Basic Cloud Knowledge

LANGUAGES

English (Fluent) · Hindi (Fluent) · Odia (Native)

HOBBIES & INTERESTS

Coding and Building Personal Projects · Learning New Technologies · Participating in Hackathons · Solving Coding Challenges · Reading Tech Blogs and Articles · Exploring AI and Machine Learning Tools · Open-Source Contributions · Playing Chess · Cooking · Listening to Music